Plant Physiology

The description of structure and variation of living organisms over a period of time ended up as two apparently irreconcilable perspectives on biology.

Two levels of organisation

- the two perspectives essentially rested on two levels of organisation of life forms and phenomena
- one described at the organismic and above level of organisation, the second at the cellular and molecular level
- the first resulted in ecology and related disciplines, the second in physiology and biochemistry

What this unit describes

- the description of physiological processes, in flowering plants as an example
- photosynthesis, respiration and ultimately plant growth and development are described in molecular terms, but in the context of cellular activities and even at organism level
- wherever appropriate, the relation of the physiological processes to environment is also discussed

In this unit

Ch 11 Photosynthesis in Higher Plants

Ch 12 Respiration in Plants

Ch 13 Plant Growth and Development

Melvin Calvin

- born in Minnesota in April, 1911, he received his Ph.D. in Chemistry from the University of Minnesota
- he served as Professor of Chemistry at the University of California, Berkeley

Radioactivity put to beneficial use

- just after world war II, when the world was under shock after the Hiroshima and Nagasaki bombings and seeing the ill-effects of radio-activity, Calvin and co-workers put radio-activity to beneficial use
- with J.A. Bassham he studied reactions in green plants forming sugar and other substances from raw materials such as carbon dioxide, water and minerals, by labelling the carbon dioxide with C14

What he proposed, and what followed

- Calvin proposed that plants change light energy to chemical energy by transferring an electron in an organised array of pigment molecules and other substances
- the mapping of the pathway of carbon assimilation in photosynthesis earned him the Nobel Prize in 1961
- the principles of photosynthesis as established by Calvin are at present being used in studies on renewable resources for energy and materials, and in basic studies in solar energy research

Autotrophs and heterotrophs

- all animals including human beings depend on plants for their food, and green plants have to synthesise the food they need
- green plants make their food through photosynthesis and are therefore called autotrophs, while all other organisms that depend on them are heterotrophs
- photosynthesis is a physico-chemical process by which green plants use light energy to drive the synthesis of organic compounds
- ultimately all living forms on earth depend on sunlight for energy, and the use of energy from sunlight by plants doing photosynthesis is the basis of life on earth

Why it matters

- it is the primary source of all food on earth
- it is also responsible for the release of oxygen into the atmosphere by green plants

11.1 What we already know

Three requirements

- simple experiments in earlier classes have shown that chlorophyll, the green pigment of the leaf, light, and CO2 are required for photosynthesis to occur
- testing a variegated leaf, or a leaf partially covered with black paper and exposed to light, for the presence of starch made it clear that photosynthesis occurred only in the green parts of the leaves in the presence of light

A candle burning in a closed bell jar

A mouse in the same closed space

A mint plant placed in the bell jar

With the plant present, the mouse stayed alive and the candle burned

Figure 11.A Priestley's experiment

The KOH experiment

- a part of a leaf is enclosed in a test tube containing KOH soaked cotton, which absorbs CO_2, while the other half is exposed to air, and the setup is placed in light for some time
- on testing the two parts for starch, the exposed part tested positive while the portion in the tube tested negative
- this showed that CO_2 was required for photosynthesis

11.2 Early experiments

Joseph Priestley, 1770

- his series of experiments revealed the essential role of air in the growth of green plants, and he discovered oxygen in 1774
- he observed that a candle burning in a closed space, a bell jar, soon gets extinguished, and that a mouse would soon suffocate in a closed space
- he concluded that a burning candle, or an animal that breathes the air, both somehow damage the air
- but when he placed a mint plant in the same bell jar the mouse stayed alive and the candle continued to burn, so he hypothesised that plants restore to the air whatever breathing animals and burning candles remove

Jan Ingenhousz

- using a setup similar to Priestley's, but placing it once in the dark and once in the sunlight, he showed that sunlight is essential for the plant process that purifies the air fouled by burning candles or breathing animals
- in an elegant experiment with an aquatic plant he showed that in bright sunlight small bubbles were formed around the green parts, while in the dark they did not
- he later identified these bubbles to be of oxygen, and so showed that it is only the green part of the plants that could release oxygen

11.3 From glucose to the balanced equation

Julius von Sachs

- it was not until about 1854 that he provided evidence for production of glucose when plants grow, glucose being usually stored as starch
- his later studies showed that the green substance in plants, chlorophyll as we know it now, is located in special bodies, later called chloroplasts, within plant cells
- he found that the green parts in plants are where glucose is made

T.W. Engelmann

- using a prism he split light into its spectral components and then illuminated a green alga, Cladophora, placed in a suspension of aerobic bacteria
- the bacteria were used to detect the sites of oxygen evolution, and he observed that they accumulated mainly in the region of blue and red light of the split spectrum
- a first action spectrum of photosynthesis was thus described, and it resembles roughly the absorption spectra of chlorophyll a and b

The empirical equation

- by the middle of the nineteenth century the key features were known, namely that plants could use light energy to make carbohydrates from CO_2 and water
- the total process for oxygen evolving organisms was understood as $CO_2 + H_2O \rightarrow [CH_2O] + O_2$ in light, where $[CH_2O]$ represented a carbohydrate such as glucose, a six-carbon sugar

Cornelius van Niel

- based on his studies of purple and green bacteria he demonstrated that photosynthesis is essentially a light-dependent reaction in which hydrogen from a suitable oxidisable compound reduces carbon dioxide to carbohydrates, expressed as $2H_2A + CO_2 \rightarrow 2A + CH_2O + H_2O$ in light
- in green plants H_2O is the hydrogen donor and is oxidised to O_2, while some organisms do not release O_2; when H_2S is the donor for purple and green sulphur bacteria, the oxidation product is sulphur or sulphate depending on the organism
- he inferred that the O_2 evolved by the green plant comes from H_2O, not from carbon dioxide, later proved by radioisotopic techniques
- the correct equation is therefore $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6H_2O + 6O_2$ in light

Outer membrane and inner membrane

Grana, and the stromal lamella connecting them

Stroma, the matrix around them

Ribosomes, starch granule and lipid droplet within

Figure 11.B Diagrammatic representation of an electron micrograph of a section of chloroplast

Not a single reaction

- the overall equation is not a single reaction but the description of a multistep process called photosynthesis

11.4 Where photosynthesis takes place

In the chloroplast

- photosynthesis takes place in the green leaves of plants, but also in other green parts
- the mesophyll cells in the leaves have a large number of chloroplasts, which usually align themselves along the walls of the mesophyll cells so that they get the optimum quantity of the incident light
- within the chloroplast there is a membranous system consisting of grana, the stroma lamellae and the matrix stroma

A division of labour

- the membrane system is responsible for trapping the light energy and also for the synthesis of ATP and NADPH
- in the stroma, enzymatic reactions synthesise sugar, which in turn forms starch
- the former set, being directly light driven, are called light reactions or photochemical reactions
- the latter are not directly light driven but depend on the products of light reactions, and by convention are called dark reactions or carbon reactions, which should not be construed to mean they occur in darkness or are not light-dependent

Chlorophyll a, with peaks in the blue and the red

Chlorophyll b, and the carotenoids alongside

Figure 11.C a Graph showing the absorption spectrum of chlorophyll a, b and the carotenoids

The wavelengths at which maximum photosynthesis occurs

Figure 11.C b Graph showing action spectrum of photosynthesis

The two curves laid over one another, showing they do not overlap completely

Figure 11.C c Graph showing action spectrum of photosynthesis superimposed on absorption spectrum of chlorophyll a

11.5 The pigments of photosynthesis

Four pigments, not one

- the colour we see in leaves is not due to a single pigment, as a chromatographic separation of leaf pigments shows
- chlorophyll a is bright or blue green in the chromatogram, chlorophyll b yellow green, xanthophylls yellow, and carotenoids yellow to yellow-orange
- pigments are substances that have an ability to absorb light at specific wavelengths

What the graphs show

- the wavelengths at which there is maximum absorption by chlorophyll a, that is in the blue and the red regions, also show a higher rate of photosynthesis
- hence we can conclude that chlorophyll a is the chief pigment associated with photosynthesis
- but the superimposed graphs show there is not a complete one-to-one overlap between the absorption spectrum of chlorophyll a and the action spectrum of photosynthesis

Pigment molecules forming the antenna

A photon absorbed and its energy funnelled inward

Reaction centre, a single chlorophyll a molecule

Primary acceptor receiving the excited electron

Figure 11.D The light harvesting complex

11.6 Accessory pigments

Widening the range

- most photosynthesis takes place in the blue and red regions of the spectrum, though some does take place at other wavelengths of the visible spectrum
- though chlorophyll a is the major pigment responsible for trapping light, other thylakoid pigments such as chlorophyll b, xanthophylls and carotenoids, called accessory pigments, also absorb light and transfer the energy to chlorophyll a
- they not only enable a wider range of wavelength to be utilised but also protect chlorophyll a from photo-oxidation

11.7 The light reaction

The photochemical phase

- light reactions include light absorption, water splitting, oxygen release, and the formation of the high-energy chemical intermediates ATP and NADPH
- the pigments are organised into two discrete light harvesting complexes within Photosystem I and Photosystem II, named in the sequence of their discovery and not in the sequence in which they function
- the complexes are made up of hundreds of pigment molecules bound to proteins, and each photosystem has all the pigments except one molecule of chlorophyll a, forming a light harvesting system also called antennae

The reaction centre

- the single chlorophyll a molecule forms the reaction centre, which is different in the two photosystems
- in PS I the reaction centre chlorophyll a has an absorption peak at 700 nm, hence is called P700
- in PS II it has absorption maxima at 680 nm, and is called P680

11.8 The electron transport

Electrons are excited

- in photosystem II the reaction centre chlorophyll a absorbs 680 nm wavelength of red light, causing electrons to become excited and jump into an orbit farther from the atomic nucleus
- these electrons are picked up by an electron acceptor which passes them to an electron transport system consisting of cytochromes

Photosystem II on the left, Photosystem I on the right

Electron acceptors above each photosystem

Electron transport system running between them

Water split at PS II, NADPH formed at the far end

Figure 11.E Z scheme of light reaction

Downhill, uphill, downhill

- the movement of electrons through the transport system is downhill in terms of an oxidation-reduction or redox potential scale, and the electrons are not used up but passed on to the pigments of photosystem I
- electrons in the reaction centre of PS I are also excited when they receive red light of wavelength 700 nm, and are transferred to another acceptor molecule of greater redox potential
- these electrons then move downhill again, this time to a molecule of energy-rich NADP+, and their addition reduces it to NADPH + H+
- this whole scheme, starting from PS II uphill to the acceptor, down the chain to PS I, excitation, transfer to another acceptor and finally downhill to NADP+, is called the Z scheme, from the shape formed when all the carriers are placed in sequence on a redox scale

11.9 Splitting of water

Where the replacement electrons come from

- the electrons moved from photosystem II must be replaced, and this is achieved by electrons available from the splitting of water
- the splitting of water is associated with PS II, water being split into 2H+, [O] and electrons, which creates oxygen, one of the net products of photosynthesis
- the reaction is $2H2O \rightarrow 4H+ + O2 + 4e-$
- the water splitting complex is associated with PS II, which itself is physically located on the inner side of the membrane of the thylakoid

Phosphorylation

- living organisms can extract energy from oxidisable substances and store it as bond energy, special substances such as ATP carrying this energy in their chemical bonds
- the process through which ATP is synthesised by cells is named phosphorylation

Photosystem I alone, with the electron cycled back

Electron transport system returning it to P700

ADP and inorganic phosphate forming ATP

No NADPH is produced by this route

Figure 11.F Cyclic photophosphorylation

11.10 Cyclic and non-cyclic photophosphorylation

Non-cyclic

- photophosphorylation is the synthesis of ATP from ADP and inorganic phosphate in the presence of light
- when the two photosystems work in a series, first PS II and then PS I, a process called non-cyclic photophosphorylation occurs
- the two photosystems are connected through an electron transport chain, as in the Z scheme, and both ATP and NADPH + H+ are synthesised by this kind of electron flow

Cyclic

- when only PS I is functional, the electron is circulated within the photosystem and phosphorylation occurs due to cyclic flow of electrons
- a possible location is the stroma lamellae, since the membranes of the grana have both PS I and PS II while the stroma lamellae lack PS II as well as the NADP reductase enzyme
- the excited electron does not pass on to NADP+ but is cycled back to the PS I complex, so cyclic flow results only in the synthesis of ATP and not of NADPH + H+
- it also occurs when only light of wavelengths beyond 680 nm is available for excitation

11.11 The chemiosmotic hypothesis

A proton gradient

- as in respiration, ATP synthesis is linked to development of a proton gradient across a membrane, here the membranes of the thylakoid
- one difference is that the proton accumulation is towards the inside of the membrane, that is in the lumen, whereas in respiration protons accumulate in the intermembrane space of the mitochondria
- since splitting of the water molecule takes place on the inner side of the membrane, the protons produced by it accumulate within the lumen of the thylakoids

Thylakoid membrane with its lumen inside

Protons accumulating in the lumen

ATP synthase spanning the membrane

ATP formed as protons move back to the stroma

Figure 11.G ATP synthesis through chemiosmosis

How the gradient is built

- as electrons move through the photosystems, protons are transported across the membrane, because the primary acceptor of electrons, located towards the outer side, transfers its electron not to an electron carrier but to an H carrier
- hence that molecule removes a proton from the stroma while transporting an electron, and when it passes its electron to the carrier on the inner side, the proton is released into the lumen
- the NADP reductase enzyme is located on the stroma side, and along with electrons from the acceptor of PS I, protons are necessary for the reduction of NADP+ to NADPH + H+, these protons also being removed from the stroma

What the gradient does

- within the chloroplast, protons in the stroma decrease in number while in the lumen there is accumulation of protons
- this creates a proton gradient across the thylakoid membrane as well as a measurable decrease in pH in the lumen
- the gradient matters because it is the breakdown of this gradient that leads to the synthesis of ATP
- the gradient is broken down due to the movement of protons across the membrane to the stroma through the transmembrane channel of the ATP synthase

11.12 ATP synthase

Two parts

- the enzyme consists of two parts, one called CF0 embedded in the thylakoid membrane, forming a transmembrane channel that carries out facilitated diffusion of protons across the membrane
- the other portion, CF1, protrudes on the outer surface of the thylakoid membrane on the side facing the stroma
- the breakdown of the gradient provides enough energy to cause a conformational change in the CF1 particle, which makes the enzyme synthesise several molecules of energy-packed ATP

What chemiosmosis requires

- a membrane, a proton pump, a proton gradient and ATP synthase
- energy is used to pump protons across a membrane, to create a gradient or a high concentration of protons within the thylakoid lumen
- ATP synthase has a channel that allows diffusion of protons back across the membrane, and this releases enough energy to activate the enzyme that catalyses the formation of ATP
- along with the NADPH produced by the movement of electrons, the ATP is used immediately in the biosynthetic reaction in the stroma, responsible for fixing CO2 and synthesis of sugars

11.13 Where the ATP and NADPH are used

The biosynthetic phase

- the products of light reaction are ATP, NADPH and O2; of these O2 diffuses out of the chloroplast while ATP and NADPH drive the processes leading to the synthesis of sugars
- this process does not directly depend on the presence of light but on the products of the light reaction, besides CO2 and H2O
- immediately after light becomes unavailable the biosynthetic process continues for some time and then stops, and if light is made available the synthesis starts again
- the first CO2 fixation product was found by Melvin Calvin, using radioactive 14C in algal photosynthesis studies, to be a 3-carbon organic acid, 3-phosphoglyceric acid or PGA, and the pathway was called the Calvin cycle after him

11.14 Two kinds of first product

C3 and C4

- experiments over a wide range of plants led to the discovery of another group where the first stable product of CO2 fixation was again an organic acid, but one with 4 carbon atoms
- this acid was identified to be oxaloacetic acid or OAA
- CO2 assimilation was thus said to be of two main types, the C3 pathway where the first product is a C3 acid, PGA, and the C4 pathway where it is a C4 acid, OAA
- these two groups of plants showed other associated characteristics

11.15 The primary acceptor of CO2

A five-carbon sugar

- the studies very unexpectedly showed that the acceptor molecule was a 5-carbon ketose sugar, ribulose bisphosphate or RuBP
- since the first product was a C3 acid, scientists believed the primary acceptor would be a 2-carbon compound, and spent many years trying to identify one before discovering the 5-carbon RuBP

11.16 The Calvin cycle

A cyclic pathway

- Calvin and his co-workers worked out the whole pathway and showed that it operated in a cyclic manner, the RuBP being regenerated
- the Calvin pathway occurs in all photosynthetic plants, whether they have C3 or C4 or any other pathways
- for ease of understanding it can be described under three stages, carboxylation, reduction and regeneration

Carboxylation

- carboxylation is the fixation of CO2 into a stable organic intermediate, and is the most crucial step of the cycle, where CO2 is utilised for the carboxylation of RuBP
- the reaction is catalysed by the enzyme RuBP carboxylase, which results in the formation of two molecules of 3-PGA
- since this enzyme also has an oxygenation activity it would be more correct to call it RuBP carboxylase-oxygenase, or RuBisCO

CO2 from the atmosphere entering at carboxylation

Ribulose-1,5-bisphosphate as the acceptor

3-phosphoglycerate formed, then reduced using ATP and NADPH

Triose phosphate leaving as sucrose and starch

Regeneration returning the acceptor so the cycle continues

Figure 11.H The Calvin cycle proceeds in three stages : (1) carboxylation, during which CO2 combines with ribulose-1,5-bisphosphate; (2) reduction, during which carbohydrate is formed at the expense of the photochemically made ATP and NADPH; and (3) regeneration during which the CO2 acceptor ribulose-1,5-bisphosphate is formed again so that the cycle continues

Reduction

- these are a series of reactions that lead to the formation of glucose
- the steps involve utilisation of 2 molecules of ATP for phosphorylation and two of NADPH for reduction, per CO2 molecule fixed
- the fixation of six molecules of CO2 and 6 turns of the cycle are required for the formation of one molecule of glucose

Regeneration

- regeneration of the CO2 acceptor molecule RuBP is crucial if the cycle is to continue uninterrupted
- the regeneration steps require one ATP for phosphorylation to form RuBP

The balance sheet

- for every CO2 molecule entering the Calvin cycle, 3 molecules of ATP and 2 of NADPH are required
- it is probably to meet this difference in the number of ATP and NADPH used in the dark reaction that cyclic phosphorylation takes place
- to make one molecule of glucose, 6 turns of the cycle are required
- in go six CO2, 18 ATP and 12 NADPH, and out come one glucose, 18 ADP and 12 NADP

11.17 The C4 pathway

What makes C4 plants special

- plants adapted to dry tropical regions have the C4 pathway, and though they have the C4 acid oxaloacetic acid as the first CO2 fixation product, they use the Calvin cycle as the main biosynthetic pathway
- they have a special type of leaf anatomy, they tolerate higher temperatures, they show a response to high light intensities, they lack a process called photorespiration, and they have greater productivity of biomass

Kranz anatomy

- the particularly large cells around the vascular bundles of C4 plants are called bundle sheath cells, and leaves with such anatomy are said to have Kranz anatomy
- Kranz means wreath, and is a reflection of the arrangement of cells
- the bundle sheath cells may form several layers around the vascular bundles, and are characterised by a large number of chloroplasts, thick walls impervious to gaseous exchange, and no intercellular spaces
- maize and sorghum are C4 plants in which the Kranz anatomy and the distribution of mesophyll cells can be observed

PEP, the primary acceptor, in the mesophyll cell

PEPcase fixing CO2 to give the C4 acid OAA

C4 acids moving to the bundle sheath cell and releasing CO2

A 3-carbon molecule returning to the mesophyll to remake PEP

The released CO2 entering the Calvin pathway

Figure 11.I Diagrammatic representation of the Hatch and Slack Pathway

The steps of the cycle

- the primary CO2 acceptor is a 3-carbon molecule, phosphoenol pyruvate or PEP, present in the mesophyll cells, and the enzyme responsible for the fixation is PEP carboxylase or PEPcase
- it is important to register that the mesophyll cells lack the RuBisCO enzyme, and the C4 acid OAA is formed in the mesophyll cells
- it then forms other 4-carbon compounds such as malic acid or aspartic acid in the mesophyll cells, which are transported to the bundle sheath cells, where these C4 acids are broken down to release CO2 and a 3-carbon molecule
- the 3-carbon molecule is transported back to the mesophyll where it is converted to PEP again, completing the cycle

Where each enzyme sits

- the CO2 released in the bundle sheath cells enters the C3 or Calvin pathway, a pathway common to all plants
- the bundle sheath cells are rich in RuBisCO but lack PEPcase, so the basic pathway that results in the formation of sugars is common to C3 and C4 plants
- the Calvin pathway occurs in all the mesophyll cells of C3 plants, but in C4 plants it does not take place in the mesophyll cells and does so only in the bundle sheath cells

11.18 Photorespiration

Why RuBisCO can bind oxygen

- the first CO2 fixation step is the reaction where RuBP combines with CO2 to form 2 molecules of 3PGA, catalysed by RuBisCO, that is RuBP + CO2 → 2 × 3PGA
- RuBisCO, the most abundant enzyme in the world, is characterised by the fact that its active site can bind to both CO2 and O2, hence the name
- it has a much greater affinity for CO2 when the CO2 and O2 are nearly equal, and the binding is competitive, so it is the relative concentration of the two that determines which will bind to the enzyme

What happens in C3 plants

- in C3 plants some O2 does bind to RuBisCO, and hence CO2 fixation is decreased
- here the RuBP, instead of being converted to 2 molecules of PGA, binds with O2 to form one molecule of phosphoglycerate and phosphoglycolate, in a pathway called photorespiration
- in this pathway there is neither synthesis of sugars nor of ATP; rather it results in the release of CO2 with the utilisation of ATP, and there is no synthesis of ATP or NADPH
- the biological function of photorespiration is not known yet

Why C4 plants avoid it

- in C4 plants photorespiration does not occur, because they have a mechanism that increases the concentration of CO2 at the enzyme site
- this takes place when the C4 acid from the mesophyll is broken down in the bundle sheath cells to release CO2, increasing the intracellular concentration of CO2
- in turn this ensures that RuBisCO functions as a carboxylase, minimising the oxygenase activity, so productivity and yields are better in these plants, which also show tolerance to higher temperatures

Table 11.A **Fill in the Columns 2 and 3 in this table to highlight the differences between C3 and C4 Plants**

Characteristics	C3 Plants	C4 Plants	Choose from
Cell type in which the Calvin cycle takes place			Mesophyll / Bundle sheath / both
Cell type in which the initial carboxylation reaction occurs			Mesophyll / Bundle sheath / both
How many cell types does the leaf have that fix CO2			Two: bundle sheath and mesophyll. One: mesophyll. Three: bundle sheath, palisade, spongy mesophyll
Which is the primary CO2 acceptor			RuBP / PEP / PGA
Number of carbons in the primary CO2 acceptor			5 / 4 / 3
Which is the primary CO2 fixation product			PGA / OAA / RuBP / PEP
Number of carbons in the primary CO2 fixation product			3 / 4 / 5
Does the plant have RuBisCO?			Yes / No / Not always
Does the plant have PEP Case?			Yes / No / Not always
Which cells in the plant have RuBisCO?			Mesophyll / Bundle sheath / none
CO2 fixation rate under high light conditions			Low / high / medium
Whether photorespiration is present at low light intensities			High / negligible / sometimes
Whether photorespiration is present at high light intensities			High / negligible / sometimes
Whether photorespiration would be present at low CO2 concentrations			High / negligible / sometimes
Whether photorespiration would be present at high CO2 concentrations			High / negligible / sometimes
Temperature optimum			30-40°C / 20-25°C / above 40°C
Examples			Cut vertical sections of leaves of different plants and observe under the microscope for Kranz anatomy, and list them in the appropriate columns

Rate of photosynthesis on the vertical axis

Light intensity on the horizontal axis

A linear rise at low intensities, marked A and B

A plateau where other factors become limiting, marked C, D and E

Figure 11.J Graph of light intensity on the rate of photosynthesis

11.19 Factors affecting photosynthesis

Internal and external

- the rate of photosynthesis is very important in determining the yield of plants including crop plants
- plant factors include the number, size, age and orientation of leaves, mesophyll cells and chloroplasts, internal CO2 concentration and the amount of chlorophyll, and these depend on the genetic predisposition and the growth of the plant
- external factors include the availability of sunlight, temperature, CO2 concentration and water, all of which affect the rate simultaneously
- though several factors interact, usually one factor is the major cause or limits the rate, so at any point the rate will be determined by the factor available at sub-optimal levels

Blackman's Law of Limiting Factors

- if a chemical process is affected by more than one factor, then its rate will be determined by the factor which is nearest to its minimal value
- it is the factor which directly affects the process if its quantity is changed
- for example, despite a green leaf and optimal light and CO2 conditions, the plant may not photosynthesise if the temperature is very low, and given the optimal temperature it will start photosynthesising

Light

- we need to distinguish between light quality, light intensity and the duration of exposure to light
- there is a linear relationship between incident light and CO2 fixation rates at low light intensities, and at higher intensities the rate does not show further increase as other factors become limiting
- light saturation occurs at 10 per cent of the full sunlight, so except for plants in shade or dense forests, light is rarely a limiting factor in nature

Too much light

- increase in incident light beyond a point causes the breakdown of chlorophyll and a decrease in photosynthesis

11.20 Carbon dioxide concentration

The major limiting factor

- the concentration of CO2 is very low in the atmosphere, between 0.03 and 0.04 per cent
- increase in concentration up to 0.05 per cent can cause an increase in CO2 fixation rates, beyond which the levels can become damaging over longer periods
- at low light conditions neither C3 nor C4 plants respond to high CO2 conditions, while at high light intensities both show an increase in the rates of photosynthesis
- C4 plants show saturation at about 360 microlitres per litre while C3 plants respond to increased CO2 and saturate only beyond 450, so current CO2 levels are limiting to C3 plants

Used in greenhouses

- the fact that C3 plants respond to higher CO2 concentration by showing increased rates of photosynthesis, leading to higher productivity, has been used for some greenhouse crops such as tomatoes and bell pepper
- they are allowed to grow in a carbon dioxide enriched atmosphere that leads to higher yields

11.21 Temperature and water

Temperature

- the dark reactions, being enzymatic, are temperature controlled, and though the light reactions are also temperature sensitive they are affected to a much lesser extent
- C4 plants respond to higher temperatures and show a higher rate of photosynthesis, while C3 plants have a much lower temperature optimum
- the temperature optimum also depends on the habitat the plant is adapted to, tropical plants having a higher optimum than plants adapted to temperate climates

Water

- even though water is one of the reactants in the light reaction, its effect as a factor is more through its effect on the plant than directly on photosynthesis
- water stress causes the stomata to close, hence reducing the CO2 availability
- water stress also makes leaves wilt, reducing the surface area of the leaves and their metabolic activity as well

11.22 Summary

The light reaction

- green plants make their own food by photosynthesis, carbon dioxide from the atmosphere being taken in by leaves through stomata and used for making carbohydrates, principally glucose and starch
- within the leaves the mesophyll cells have a large number of chloroplasts responsible for CO2 fixation, the membranes being sites for the light reaction while the chemosynthetic pathway occurs in the stroma
- in the light reaction the light energy is absorbed by the pigments in the antenna and funnelled to special chlorophyll a molecules called reaction centre chlorophylls, PS I having a P700 molecule and PS II a P680 reaction centre
- after absorbing light, electrons are excited and transferred through PS II and PS I and finally to NAD, forming NADH; a proton gradient is created across the thylakoid membrane, and its breakdown through the ATPase enzyme releases enough energy for synthesis of ATP

The carbon fixation cycle

- splitting of water molecules is associated with PS II, resulting in release of O2, protons and transfer of electrons
- in the carbon fixation cycle CO2 is added by RuBisCO to the 5-carbon compound RuBP, which is converted to 2 molecules of 3-carbon PGA, then to sugar by the Calvin cycle, with RuBP regenerated and the ATP and NADPH from the light reaction utilised
- RuBisCO also catalyses a wasteful oxygenation reaction in C3 plants, photorespiration
- some tropical plants show the C4 pathway, in which the first product of CO2 fixation in the mesophyll is a 4-carbon compound, the Calvin pathway then being carried out in the bundle sheath cells

11.23 Exercises

From this chapter

1. By looking at a plant externally, can you tell whether a plant is C3 or C4? Why and how?
2. By looking at which internal structure of a plant can you tell whether it is C3 or C4? Explain.
3. Even though very few cells in a C4 plant carry out the biosynthetic Calvin pathway, they are highly productive. Can you discuss why?

From this chapter, continued

1. RuBisCO is an enzyme that acts both as a carboxylase and an oxygenase. Why do you think RuBisCO carries out more carboxylation in C4 plants?
2. Suppose there were plants that had a high concentration of chlorophyll b but lacked chlorophyll a, would they carry out photosynthesis? Then why do plants have chlorophyll b and other accessory pigments?
3. Why does the colour of a leaf kept in the dark frequently become yellow, or pale green? Which pigment do you think is more stable?
4. Look at leaves of the same plant on the shady side and compare them with the leaves on the sunny side, or compare potted plants kept in the sunlight with those in the shade. Which of them has leaves that are darker green, and why?
5. Figure 11.J shows the effect of light on the rate of photosynthesis. Based on the graph: (a) at which points in the curve is light a limiting factor? (b) what could be the limiting factors in region A? (c) what do C and D represent on the curve?
6. Give a comparison between the following: (a) C3 and C4 pathways (b) cyclic and non-cyclic photophosphorylation (c) anatomy of leaf in C3 and C4 plants